

# Strategic Fantasy Baseball Draft Optimization

Building a Scalable Decision-Support System for Front Offices

OR114-2 Final Project, Group 4

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Spring 2026

## Abstract

For a modern Major League Baseball (MLB) franchise, the draft is the most cost-effective avenue for building a championship roster. As internal scouting and player evaluation models converge across the league, the competitive edge shifts from merely identifying talent to executing a mathematically superior drafting strategy. In this report, we model the draft as an operations research problem. Because the pool of draft-eligible players is massive, traditional Integer Linear Programming (ILP) fails to scale. We propose a highly efficient heuristic, *Opportunity Cost Greedy (OCG)*, which estimates the delay-cost of waiting for future turns. Tested across various synthetic market stressors and real-world 2026 projections, OCG delivers near-optimal rosters in a fraction of a second, effectively bridging the gap between exact optimization and massive-scale practicality.

## 1 Introduction: The Front Office Perspective

The origins of Fantasy Baseball (FB) can be traced back to the 1960s, when John Burgeson developed the first computer-simulated baseball game using an IBM 1620. However, the modern strategic framework of the game was primarily shaped in the 1980s by writer Daniel Okrent. Okrent’s “Rotisserie” system transformed player evaluation into a competitive management simulation. Today, FB has evolved into a sophisticated strategic game where participants act as General Managers (GMs), constructing virtual teams whose success depends on the real-life statistical performance of MLB players throughout a 162-game season. In the analytically driven Head-to-Head (H2H) Points format, diverse statistical categories are translated into a single scalar value via a predetermined weighting system, turning the construction of a roster into a complex portfolio optimization race.

For any MLB front office, constructing a roster before the season begins is the ultimate optimization puzzle. The most efficient and critical way to acquire talent is through the draft. Every team possesses internal scouting models and proprietary projections (e.g., WAR, fantasy points) to evaluate players. However, in the modern data-driven era, these evaluations often converge. When all general managers have access to similar player rankings, the true competitive advantage is no longer just *who* you evaluate as good, but *when* and *how* you draft them relative to your opponents and your roster constraints.

While the actual MLB draft uses a fixed order based on reverse standings, we abstract our environment into a **Snake Draft**. This format serves as a perfect proxy to simulate a highly competitive, fair, and strategically demanding environment. In a snake draft, a manager owns a fixed sequence of picks, and the player pool is continually depleted by opponents. The practical decision is not simply selecting the player with the highest isolated score (the naive “Best Player Available” approach).

Instead, a manager must answer a harder question: *Which player must I draft now because market timing and positional scarcity will make equivalent options disappear by my next turn?*

This decision is inherently interdependent. A high-projection player might be suboptimal if he occupies a position where supply is abundant, whereas a lower-projection player might be highly valuable if he fills a scarce roster slot. To solve this, we translate the draft into a mathematical model.

Initially, an Average Draft Position (ADP)-aware **Integer Linear Program (ILP)** is formulated. While ILP provides a flawless mathematical optimum, it encounters a severe roadblock in reality. Because baseball players do not need to actively register for the draft (anyone meeting age/school requirements is eligible), the actual draft pool scales to tens of thousands of players. When the problem scale rises, the ILP algorithm exceeds reasonable time limits or crashes completely due to exponential complexity.

To solve this, we introduce a self-designed algorithm: **Opportunity Cost Greedy (OCG)**. This heuristic explicitly calculates the mathematical cost of waiting by peeking into the future availability of players. The overarching narrative of this report is to prove to the front office that OCG can achieve incredibly tight optimality gaps (retaining maximum roster value) while executing in mere seconds, even when the player pool scales to massive extremes.

## 2 Problem Description

We consider a snake draft environment where a manager operates a predetermined set of picks. At each owned pick, the manager must choose exactly one available player from the remaining pool. Every player in the pool possesses three attributes:

1. **Projected Value:** The expected statistical output (points) for the season.
2. **Eligible Positions:** The real-life positions the player can occupy.
3. **Average Draft Position (ADP):** A market consensus of when the player is expected to be drafted by opponents.

The objective is to maximize the sum of projected points for the starting roster. The drafted players must perfectly fulfill strict positional requirements. The base roster structure used in our study demands 16 active slots, as detailed in Table 1. (Note: The Utility (**Util**) slot is hitter-only, meaning pitchers cannot be assigned there).

Table 1: Base fantasy baseball roster requirements.

| Position       | C | 1B | 2B | 3B | SS | OF | Util | SP | RP |
|----------------|---|----|----|----|----|----|------|----|----|
| Required Slots | 1 | 1  | 1  | 1  | 1  | 3  | 1    | 5  | 2  |

A player can only be drafted if he is still available in the market. We treat ADP as a proxy for market depletion. If the manager’s current draft pick occurs much later than a player’s ADP, that player is considered successfully drafted by an opponent and is unavailable. By integrating ADP market constraints and positional flexibility into the sequence of decisions, draft planning is transformed from a subjective ranking exercise into a strict operations research problem.

### 3 Model Formulation

We first formulate the exact ADP-aware Integer Linear Program (ILP) to serve as our absolute benchmark for roster quality.

Let  $I$  be the set of players,  $P$  the set of roster positions, and  $K$  the set of owned draft picks. Each player  $i \in I$  has projected points  $v_i$ , market expectation  $\text{adp}_i$ , and eligible positions  $E_i \subseteq P$ . Each position  $p \in P$  requires  $r_p$  players. Let  $S_k$  be the overall draft number for the manager’s  $k$ -th pick.

**Decision Variables:**

- $y_i \in \{0, 1\}$ : 1 if player  $i$  is drafted, 0 otherwise.
- $x_{ip} \in \{0, 1\}$ : 1 if player  $i$  is assigned to position  $p$ , 0 otherwise.
- $z_{ik} \in \{0, 1\}$ : 1 if player  $i$  is selected precisely at owned pick  $k$ , 0 otherwise.

**Objective Function:** Maximize total roster points.

$$\max \sum_{i \in I} v_i y_i$$

**Constraints:**

$$\sum_{i \in I} z_{ik} = 1, \quad \forall k \in K \quad (1)$$

$$\sum_{k \in K} z_{ik} = y_i, \quad \forall i \in I \quad (2)$$

$$\sum_{p \in P} x_{ip} = y_i, \quad \forall i \in I \quad (3)$$

$$x_{ip} \leq 1, \quad \forall i \in I, p \in E_i \quad (4)$$

$$\sum_{i \in I} x_{ip} = r_p, \quad \forall p \in P \quad (5)$$

$$z_{ik} = 0 \quad \text{if } S_k > \text{adp}_i + \delta, \quad \forall i \in I, k \in K \quad (6)$$

$$y_i, x_{ip}, z_{ik} \in \{0, 1\} \quad \forall i, p, k \quad (7)$$

Constraint (1) enforces exactly one selection per pick. (2) and (3) ensure logical consistency between picking, drafting, and assigning. (4) restricts assignments to legal positions. (5) dictates the roster shape. Constraint (6) is the vital **Market Availability Constraint**: a player cannot be drafted if the current pick  $S_k$  is later than his expected depletion point (ADP + a tolerance buffer  $\delta$ ).

### 4 Algorithms

While the IP provides mathematical perfection, solving it is NP-Hard. For massive player pools, the branch-and-bound tree explodes. Therefore, we developed two heuristic algorithms using Priority Queues (Heaps) to achieve extreme computational efficiency.

#### 4.1 Direct Greedy (Baseline)

The Direct Greedy (DG) algorithm serves as our baseline. It represents the standard “fill the biggest need with the best available player” mentality.

At each pick, DG scans all open positions, identifies the position with the highest scarcity (Remaining Need / Available Players), and selects the player with the highest projected points for that position.

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**Algorithm 1** Direct Greedy (DG)

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1: Initialize Max-Heaps for each position  $p \in P$  based on  $v_i$ .
2: Sort all players by  $(\text{adp}_i + \delta)$  to create an expiration_order.
3: for each owned pick  $k \in K$  do
4:   Pop expired players from heaps where  $\text{adp}_i + \delta < S_k$ 
5:    $\text{best\_position} \leftarrow \text{null}$ ,  $\text{max\_scarcity} \leftarrow -\infty$ 
6:   for each position  $p$  with  $r_p > 0$  do
7:     Clean heap (remove drafted/expired players lazily)
8:      $\text{scarcity} \leftarrow \frac{\text{Remaining Need for } p}{\text{Active Eligible Players for } p}$ 
9:     if  $\text{scarcity} > \text{max\_scarcity}$  (tie-break by top player's  $v_i$ ) then
10:       $\text{best\_position} \leftarrow p$ 
11:       $\text{max\_scarcity} \leftarrow \text{scarcity}$ 
12:     end if
13:   end for
14:   Draft top player from  $\text{best\_position}$  heap and update remaining needs.
15: end for

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## 4.2 Opportunity Cost Greedy (Proposed Method)

*Opportunity Cost Greedy (OCG)* is our flagship algorithm. Instead of only looking at the *current* turn, OCG looks ahead to the *next* owned pick. It calculates the **Opportunity Cost** of waiting for a position by subtracting the points of the best player available *next round* from the best player available now.

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**Algorithm 2** Opportunity Cost Greedy (OCG)

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1: Initialize two sets of Max-Heaps for each position  $p$ : current_heaps and future_heaps.
2: for each owned pick  $k \in K$  (let next pick be  $k_{\text{next}}$ ) do
3:   Expire players in current_heaps based on  $S_k$ 
4:   Expire players in future_heaps based on  $S_{k_{\text{next}}}$ 
5:    $\text{best\_choice} \leftarrow \text{null}$ ,  $\text{max\_priority} \leftarrow (-\infty, -\infty, -\infty)$ 
6:   for each position  $p$  with  $r_p > 0$  do
7:      $v_{\text{cur}} \leftarrow$  points of top player in current_heaps[ $p$ ]
8:      $v_{\text{fut}} \leftarrow$  points of top player in future_heaps[ $p$ ]
9:      $\text{score} \leftarrow v_{\text{cur}} - v_{\text{fut}}$  ▷ Delay Cost Calculation
10:    forced  $\leftarrow$  True if (Active Players  $\leq$  Need)
11:    priority  $\leftarrow$  (forced,  $\text{score}$ ,  $v_{\text{cur}}$ )
12:    if priority  $>$   $\text{max\_priority}$  then
13:       $\text{max\_priority} \leftarrow$  priority
14:       $\text{best\_choice} \leftarrow$  (Top Player,  $p$ )
15:    end if
16:  end for
17:  Draft player from  $\text{best\_choice}$  and update remaining needs.
18: end for

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## 4.3 Time Complexity Analysis

Let  $n$  be the total number of players in the draft pool,  $r$  be the number of roster slots (total picks per manager), and  $p$  be the number of distinct positions.

- **Initialization:** Building the initial max-heaps and sorting the expiration list takes  $O(n \log n)$ .
- **Drafting Loop:** Checking the top of a heap is  $O(1)$ . For each of the  $r$  picks, the algorithm scans  $p$  positions.
- **Heap Cleaning (Lazy Deletion):** Expiring or removing drafted players is done by popping invalid elements at the top of the heap. Across the *entire draft process*, a player is popped from a position heap at most once. Therefore, the total time spent cleaning heaps is bounded by  $O(n \log n)$ .

**Total Time Complexity for both DG and OCG is  $O(n \log n + r \cdot p)$ .** Because  $r$  and  $p$  are tiny constants relative to  $n$ , the algorithm runs in near-linear time relative to sorting the player pool. This mathematically guarantees extreme scalability compared to the exponential worst-case nature of IP.

## 5 Data Collection and Generation

To evaluate our algorithms, we utilize both real-world data and a highly customizable **Synthetic Data Generator**.

### 5.1 Real Data Collection

Real data validation serves as our first step of validation. We constructed our real dataset by downloading the Average Draft Position (ADP) and 2026 player projected batting and pitching statistics from FantasyPros. Figures 1 and 2 display examples of the projected data and ADP metrics collected.

| PLAYER                                    | AB  | R   | HR | RBI | SB | AVG  | OBP  | H   | 2B | 3B | BB  | SO  | SLG  | OPS   | ROST% |
|-------------------------------------------|-----|-----|----|-----|----|------|------|-----|----|----|-----|-----|------|-------|-------|
| <b>Shohei Ohtani</b><br>(LAD - SP,DH)     | 572 | 124 | 47 | 111 | 24 | .283 | .382 | 162 | 28 | 5  | 92  | 164 | .598 | .980  | 100%  |
| <b>Aaron Judge</b><br>(NYY - LF,CF,RF,DH) | 520 | 110 | 46 | 114 | 9  | .292 | .417 | 152 | 26 | 1  | 112 | 160 | .609 | 1.026 | 99%   |
| <b>Bobby Witt Jr.</b><br>(KC - SS)        | 603 | 101 | 29 | 93  | 34 | .292 | .345 | 176 | 38 | 6  | 49  | 114 | .518 | .863  | 99%   |

Figure 1: Example of 2026 player projected batting and pitching statistics.

| Import a Team |                                          | 2026  |     |     |      |    |      |     |
|---------------|------------------------------------------|-------|-----|-----|------|----|------|-----|
| Rank          | Player (Team)                            | Yahoo | CBS | RTS | NFBC | FT | ESPN | AVG |
| 1             | <b>Shohei Ohtani</b> (LAD - SP,DH)       |       | 1   | 2   | 1    | 1  | 1    | 1.0 |
| 2             | <b>Aaron Judge</b> (NYY - LF,CF,RF,DH)   | 1     | 2   | 1   | 2    | 2  | 2    | 1.8 |
| 3             | <b>Shohei Ohtani (Batter)</b> (LAD - DH) | 2     |     |     |      |    |      | 2.0 |
| 4             | <b>Bobby Witt Jr.</b> (KC - SS)          | 3     | 3   | 4   | 3    | 4  | 4    | 3.6 |
| 5             | <b>Juan Soto</b> (NYM - LF,RF,DH)        | 4     | 4   | 3   | 4    | 3  | 3    | 3.6 |

Figure 2: Example of player Average Draft Position (ADP).

Player values are calculated using two distinct standard scoring benchmarks: Yahoo (which focuses on traditional counting stats) and FanGraphs (which incorporates more advanced player metrics based on linear weights). The detailed scoring weights for both batters and pitchers are outlined in Table 2.

Table 2: Scoring weights for Yahoo and FanGraphs formats.

| Metric    | Batters |           | Metric | Pitchers |           |
|-----------|---------|-----------|--------|----------|-----------|
|           | Yahoo   | FanGraphs |        | Yahoo    | FanGraphs |
| 1B (or H) | 2.6     | 5.6       | IP     | 3.0      | 7.4       |
| 2B        | 5.2     | 2.9       | K      | 3.0      | 2.0       |
| 3B        | 7.8     | 5.7       | W      | 8.0      | —         |
| HR        | 10.4    | 9.4       | SV     | 8.0      | 5.0       |
| R         | 1.9     | —         | ER     | -3.0     | —         |
| RBI       | 1.9     | —         | BB     | -1.3     | -3.0      |
| BB        | 2.6     | 3.0       | H      | -1.3     | -2.6      |
| SB        | 4.2     | 1.9       | HBP    | -1.3     | -3.0      |
| HBP       | 0 (2.6) | 0 (3.0)   | HR     | —        | -12.3     |
| AB        | —       | -1.0      | HLD    | —        | 4.0       |
| CS        | —       | 0 (-2.8)  |        |          |           |

Slight modifications were made to the original projected stats to align with these scoring categories. For example, since the projected stats lack Hit By Pitch (HBP) and Caught Stealing (CS) for batters, we effectively set their scoring weights to zero (original standard weights are shown in parentheses in the table). Furthermore, since outs are required for pitching calculations, we compute them as Innings Pitched (IP)  $\times$  3.

Additionally, across all experiments (both real and synthetic data), the manager’s initial draft position is standardized to the middle of the draft order (e.g., the 6th pick in a 12-team league, or the 12th pick in a 24-team league) to maintain a consistent evaluation baseline.

## 5.2 Synthetic Data Generation

While real data confirms practical viability, the most important and further validation is performed on synthetic data. Synthetic data is indispensable as it allows us to mathematically isolate specific operational variables (Factors)—such as talent distribution, positional versatility, and market supply—and stress-test the algorithms under controlled extreme conditions.

The total number of generated players in the synthetic environment is defined dynamically based on the league size ( $T$ ), base roster slots ( $R = 16$ ), roster scale multiplier ( $S$ ), and player-demand ratio ( $D$ ):

$$N_{\text{players}} = S \times 16 \times T \times D$$

Our experimental framework generates synthetic profiles encompassing four primary operational factors:

## 5.3 Factor A: Points Distribution ( $v_i$ )

We designed three points distributions to simulate different talent pools, clipping all generated values to a realistic range of  $[10.0, 800.0]$  according to the real data we collected above:

- **Normal (Baseline):** Values follow a normal distribution  $v_i \sim \mathcal{N}(400, 120^2)$ . This represents an average draft class where the majority of players are solid, mid-tier contributors.
- **Uniform:** Values are uniformly distributed  $v_i \sim \mathcal{U}(10, 800)$ . This flattens the talent curve, offering many viable replacement options and minimizing scarcity.
- **High-Low (Star-Heavy):** Simulates a highly skewed draft class. Only 10% of players are classified as elites, with values drawn from  $\mathcal{U}(500, 800)$ . The remaining 90% fall into  $\mathcal{U}(10, 500)$ . In this scenario, missing out on top-tier talent is severely punished in the objective function.

## 5.4 Factor B: Position Eligibility Distribution ( $E_i$ )

We control how versatile the player pool is using four distinct generation logics:

- **Roster-Ratio (Baseline):** Players are assigned exactly one position, but the probability of generating a specific position aligns precisely with roster demand (e.g., more OF and SP players are generated).
- **Uniform-by-Type:** Randomly designates a player as a hitter or pitcher, then uniformly assigns a position pattern (including dual-eligibility like 1B;3B or SP;RP).
- **Point-Flexible:** The most forgiving environment. The generator ranks all players by points. Players in the top 20% (elite) have an 80% chance of gaining multi-position eligibility (up to 3 positions). This mimics real life where superstars often possess high versatility.
- **Single-Position:** Every player possesses exactly one eligible position. This eliminates all flexibility, representing the ultimate test of scarcity and making early greedy mistakes almost impossible to repair.

## 5.5 Factor C: Market Volatility and ADP Noise ( $\sigma_{\text{adp}}$ )

To generate ADP, we first sort all players by their true projected points in descending order to establish a “True Rank”. We then simulate market noise by adding a Gaussian error term:

$$\text{ADP}_i = \text{TrueRank}_i + \mathcal{N}(0, \sigma_{\text{adp}}^2)$$

- $\sigma_{\text{adp}} = 0$  represents a perfectly efficient market where opponents draft strictly by objective value.
- Higher  $\sigma_{\text{adp}}$  introduces market chaos. Opponents may make unpredictable “reaches” or leave bargains on the board. This tests how robust our algorithms are when future availability is uncertain.

## 5.6 Factor D: Player Demand Ratio ( $D$ )

To test the algorithms under varying levels of scarcity, we manipulate the Player Demand Ratio ( $D$ ). This factor controls the supply-and-demand economics of the draft pool by dictating the ratio of “Available Players” versus “Total League Demand”. Let  $T$  be the number of teams and  $R = 16$  be the base roster slots. The total number of generated players is strictly governed by  $N_{\text{players}} = R \times T \times D$ .

- $D = 1$  (**Extreme Scarcity**): The league will draft almost every available player. One greedy mistake means replacing a starter with absolute zero value.
- $D = 3$  (**Baseline Supply**): Represents standard fantasy draft conditions with a typical buffer of undrafted free agents.
- $D = 10$  (**High Abundance**): The player pool is a massive ocean of options.

## 5.7 Scenario Matrix

We developed 15 distinct scenarios to isolate the impact of each factor. As shown in Table 3, **Scenario S1** serves as the baseline for all comparative analyses. The remaining scenarios follow the factor order introduced above: projected points distribution, position eligibility distribution, market volatility, and player demand ratio. For every scenario, we generate **30 random instances** to calculate the mean ( $\mu = 0$ ) and standard deviation ( $\sigma$ ) of the performance metrics.

Table 3: Experimental Scenario Matrix for Draft Optimization

| ID      | Main Factor          | Demand Ratio ( $D$ ) | Points Dist. ( $v_i$ ) | Pos. Dist. ( $E_i$ )   | ADP Noise ( $\delta, \sigma$ ) |
|---------|----------------------|----------------------|------------------------|------------------------|--------------------------------|
| S1      | <b>Baseline</b>      | 3                    | Normal                 | Roster-Ratio           | (0, 30)                        |
| S2      | Points: Uniform      | 3                    | <b>Uniform</b>         | Roster-Ratio           | (0, 30)                        |
| S3      | Points: Star-Heavy   | 3                    | <b>High-Low</b>        | Roster-Ratio           | (0, 30)                        |
| S4      | Pos: Uniform-by-Type | 3                    | Normal                 | <b>Uniform-by-Type</b> | (0, 30)                        |
| S5      | Pos: Point-Flexible  | 3                    | Normal                 | <b>Point-Flexible</b>  | (0, 30)                        |
| S6      | Pos: Single-Position | 3                    | Normal                 | <b>Single-Position</b> | (0, 30)                        |
| S7      | Market: Efficient    | 3                    | Normal                 | Roster-Ratio           | (0, 0)                         |
| S8      | Market: Mild Noise   | 3                    | Normal                 | Roster-Ratio           | (0, 60)                        |
| S9, 10  | Market: Chaotic      | 3                    | Normal                 | Roster-Ratio           | (±5, 30)                       |
| S11, 12 | Market: Chaotic      | 3                    | Normal                 | Roster-Ratio           | (±10, 30)                      |
| S13     | Load (High)          | 1                    | Normal                 | Roster-Ratio           | (0, 30)                        |
| S14     | Load (Low)           | 3                    | Normal                 | Roster-Ratio           | (0, 30)                        |
| S15     | Load (Ultra-Low)     | 10                   | Normal                 | Roster-Ratio           | (0, 30)                        |

## 5.8 Stress Testing Design

To evaluate the absolute computational limits of our algorithms relative to the ILP benchmark, we design a separate **Large-Scale Stress Test** outside of the standard factor matrix. Instead of sweeping isolated market parameters, this test scales the raw problem dimensions—drastically increasing both the available player pool and the roster constraints. The primary objective is to simulate massive, unconstrained draft databases to observe the breaking point of exact optimization (ILP) and to validate the asymptotic time efficiency of the OCG heuristic under massive scale.

## 6 Performance Evaluation

Algorithms are evaluated using two primary metrics across 30 random seeds for each scenario:

- **Optimal Gap Ratio (%)**: The distance from the ILP benchmark, defined as  $\frac{Z_{ILP} - Z_{Heuristic}}{Z_{ILP}} \times 100\%$ . A lower value indicates higher accuracy.
- **Improvement (%)**: The direct gain of OCG over the DG baseline, calculated per seed as  $\frac{Z_{OCG} - Z_{DG}}{Z_{DG}} \times 100\%$  and reported as  $\mu \pm \sigma$ .

Unless otherwise noted, all synthetic results are reported as Mean  $\pm$  Standard Deviation.

### 6.1 Real-Data Validation

As the first step of validation, we ran the algorithms on the real-world 2026 Yahoo and FanGraphs projections. As seen in Table 4, OCG successfully trims the point loss caused by naive greedy drafting, confirming its practical viability on real MLB data structures before proceeding to the more comprehensive synthetic validations.

Table 4: Real-data validation (Gap measured in optimal gap ratio versus ILP).

| Scoring Method | Method                  | Optimal Gap Ratio |
|----------------|-------------------------|-------------------|
| Yahoo 2026     | Opportunity Cost Greedy | <b>0.50%</b>      |
|                | Direct Greedy           | 3.24%             |
| FanGraphs 2026 | Opportunity Cost Greedy | <b>1.24%</b>      |
|                | Direct Greedy           | 3.36%             |

## 6.2 Factor A: Points Distribution

Table 5 summarizes the impact of the talent distribution. Under the **Baseline (Normal)** distribution, OCG achieves a tight gap of 1.92%, reducing the error of the DG baseline by more than half. The advantage is most pronounced in **Star-Heavy** markets (S3), where OCG prevents elite talent from being snatched by opponents, yielding a **6.10%** average improvement in total roster value.

Table 5: Results for Factor A: Points Distribution. Values represent the optimality gap (%)  $\pm$  standard deviation.

| Level                 | DG Gap (%)        | OCG Gap (%)                     | Improvement (%)                 |
|-----------------------|-------------------|---------------------------------|---------------------------------|
| Normal (Baseline)     | 4.79% $\pm$ 1.18% | <b>1.92%</b> $\pm$ <b>0.57%</b> | 3.02% $\pm$ 1.34%               |
| Uniform               | 3.24% $\pm$ 0.77% | <b>1.44%</b> $\pm$ <b>0.59%</b> | 1.87% $\pm$ 0.65%               |
| High-Low (Star-Heavy) | 8.39% $\pm$ 2.40% | <b>2.87%</b> $\pm$ <b>1.32%</b> | <b>6.10%</b> $\pm$ <b>3.46%</b> |

## 6.3 Factor B: Position Eligibility Distribution

Table 6 illustrates performance across different positional flexibility environments. Compared to the **Baseline (Roster-Ratio)**, *point-flexible* is the most forgiving environment (gap drops to 1.34%) because top players can seamlessly plug holes in the roster. Conversely, *single-position* creates a rigid, unforgiving draft. Even when mistakes are impossible to patch with utility players, OCG dominates Direct Greedy (0.94% vs 5.11%), proving that looking one pick ahead prevents catastrophic positional deficits later in the draft.

Table 6: Results for Factor B: Position Eligibility Distribution. Values represent the optimality gap (%)  $\pm$  standard deviation.

| Level                   | DG Gap (%)        | OCG Gap (%)                     | Improvement (%)                 |
|-------------------------|-------------------|---------------------------------|---------------------------------|
| Roster-Ratio (Baseline) | 4.79% $\pm$ 1.18% | <b>1.92%</b> $\pm$ <b>0.57%</b> | 3.02% $\pm$ 1.34%               |
| Uniform-by-Type         | 4.89% $\pm$ 0.62% | <b>1.07%</b> $\pm$ <b>0.43%</b> | 4.02% $\pm$ 0.82%               |
| Point-Flexible          | 5.36% $\pm$ 1.66% | <b>1.34%</b> $\pm$ <b>0.68%</b> | 4.27% $\pm$ 1.83%               |
| Single-Position         | 5.11% $\pm$ 1.48% | <b>0.94%</b> $\pm$ <b>0.65%</b> | <b>4.42%</b> $\pm$ <b>1.95%</b> |

## 6.4 Factor C: Market Volatility and ADP Noise

Table 7 demonstrates performance under chaotic draft markets. We split the analysis into two sweeps: varying market noise ( $\sigma_{\text{adp}}$ ) and varying availability tolerance ( $\delta$ ).

When the market is perfectly efficient ( $\sigma_{\text{adp}} = 0$ ), predicting opponent behavior is trivial, and both algorithms perform exceptionally well. As noise increases to  $\sigma_{\text{adp}} = 60$ , the draft becomes unpredictable. Direct Greedy struggles significantly (gap over 5%), whereas OCG remains highly robust (2.43%). Similarly, whether the availability tolerance ( $\delta$ ) is incredibly strict ( $-10$ ) or highly permissive ( $10$ ), OCG consistently keeps the gap under 2%. Because OCG bases its priority on the *relative difference* (Opportunity Cost) rather than absolute values, it inherently hedges against market volatility.

Table 7: Factor C: ADP Uncertainty results. The first sweep tests market noise ( $\sigma_{\text{adp}}$ ) with fixed  $\delta = 0$ . The second tests tolerance ( $\delta$ ) with fixed  $\sigma_{\text{adp}} = 30$ . Values represent the optimality gap (%)  $\pm$  standard deviation.

| Sweep            | Level                                 | DG Gap (%)        | OCG Gap (%)                     | Improvement (%)                 |
|------------------|---------------------------------------|-------------------|---------------------------------|---------------------------------|
| <b>Noise</b>     | $\sigma_{\text{adp}} = 0$ (Efficient) | 1.47% $\pm$ 0.48% | <b>0.48%</b> $\pm$ <b>0.23%</b> | 1.00% $\pm$ 0.50%               |
|                  | $\sigma_{\text{adp}} = 30$ (Moderate) | 4.79% $\pm$ 1.18% | <b>1.92%</b> $\pm$ <b>0.57%</b> | 3.02% $\pm$ 1.34%               |
|                  | $\sigma_{\text{adp}} = 60$ (Chaotic)  | 5.56% $\pm$ 1.58% | <b>2.43%</b> $\pm$ <b>1.55%</b> | <b>3.33%</b> $\pm$ <b>1.43%</b> |
| <b>Tolerance</b> | $\delta = -10$ (Strict)               | 4.65% $\pm$ 1.32% | <b>1.90%</b> $\pm$ <b>0.77%</b> | 2.91% $\pm$ 1.71%               |
|                  | $\delta = 0$ (Neutral)                | 4.79% $\pm$ 1.18% | <b>1.92%</b> $\pm$ <b>0.57%</b> | 3.02% $\pm$ 1.34%               |
|                  | $\delta = 10$ (Permissive)            | 5.42% $\pm$ 1.21% | <b>1.72%</b> $\pm$ <b>0.95%</b> | <b>3.92%</b> $\pm$ <b>1.03%</b> |

## 6.5 Factor D: Player Demand Ratio

Table 8 highlights scaling under tight vs. loose supply (Player Demand Ratio  $D$ ). When supply is brutally tight ( $D = 1$ , meaning almost every player must be drafted), Direct Greedy collapses (8.81% gap) as it blindly paints itself into a corner. OCG, aware of the impending scarcity, navigates the tight market with only a 2.64% gap. In a high-abundance scenario ( $D = 10$ ), OCG drops the error rate to a mere 1.16%.

Table 8: Factor D: Player Demand Ratio results (Fixed Roster Scale = 1). Values represent the optimality gap (%)  $\pm$  standard deviation.

| Level                    | DG Gap (%)        | OCG Gap (%)                     | Improvement (%)                 |
|--------------------------|-------------------|---------------------------------|---------------------------------|
| D = 1 (Extreme Scarcity) | 8.81% $\pm$ 2.19% | <b>2.64%</b> $\pm$ <b>1.23%</b> | <b>6.82%</b> $\pm$ <b>3.13%</b> |
| D = 3 (Baseline Supply)  | 4.79% $\pm$ 1.18% | <b>1.92%</b> $\pm$ <b>0.57%</b> | 3.02% $\pm$ 1.34%               |
| D = 10 (High Abundance)  | 3.29% $\pm$ 0.85% | <b>1.16%</b> $\pm$ <b>0.51%</b> | 2.20% $\pm$ 0.83%               |

## 6.6 Large-Scale Stress Testing

This section presents the most critical argument for our front office pitch: **Scalability**. While the ILP finds perfect rosters in smaller instances, what happens when the scouting department dumps a universe of hundreds of thousands of eligible players into the database?

Table 9 showcases the **Large-Scale Stress Test**. To understand the complexity of the ILP, we approximate its search space by the variable count. For  $n$  available players,  $p$  roster positions, and a roster size of  $r$ , the approximate number of decision variables is:

$$\text{Approx. Variables} = n \times (1 + p + r)$$

This formula encompasses the player-selection variables ( $y_i$ ), player-position assignment variables ( $x_{ip}$ ), and player-round availability variables ( $z_{ik}$ ). As we scale the roster size to 256 and the player pool to 184,320, the approximate variable count reaches nearly 50 million.

The Branch-and-Bound tree of the ILP explores an exponentially growing space. As a result, the ILP runtime explodes from 4 seconds to 276 seconds, and eventually hits our predefined 30-minute (1800 seconds) solver time limit for the largest target, triggering a `TIMEOUT_ERROR`. In stark contrast, confirming our  $O(n \log n)$  complexity analysis, **Opportunity Cost Greedy processes the massive 50-million variable equivalent environment in just 23.286 seconds.**

Table 9: Stress Test: Execution Times for DG, OCG, and ILP.

| Instance Level | Players | DG Time(s) | OCG Time(s) | ILP Time(s)     | ILP Status     |
|----------------|---------|------------|-------------|-----------------|----------------|
| stress_small   | 5,760   | 0.790      | 0.852       | 4.235           | OPTIMAL        |
| stress_medium  | 9,600   | 0.951      | 1.144       | 12.230          | OPTIMAL        |
| stress_large   | 21,600  | 1.981      | 2.344       | 54.267          | OPTIMAL        |
| stress_xlarge  | 64,000  | 6.172      | 10.360      | 276.895         | OPTIMAL        |
| timeout_target | 184,320 | 17.760     | 23.286      | <b>1869.805</b> | <b>TIMEOUT</b> |

Table 10: Stress Test: DG and OCG Optimality Gaps Relative to ILP.

| Instance Level | DG Gap (%) | OCG Gap (%)   |
|----------------|------------|---------------|
| stress_small   | 2.430%     | <b>1.098%</b> |
| stress_medium  | 1.771%     | <b>0.646%</b> |
| stress_large   | 0.940%     | <b>0.407%</b> |
| stress_xlarge  | 0.270%     | <b>0.137%</b> |
| timeout_target | —          | —             |

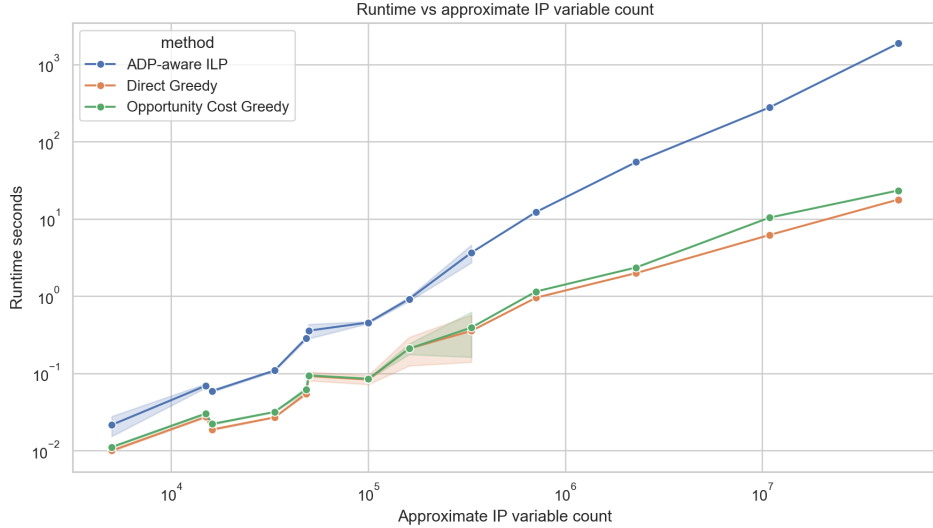


Figure 3: Runtime comparison for Stress Testing. ILP scales exponentially, while heuristics scale linearly.

## 6.7 Strategic Insights and Competitive Edge

After evaluating the algorithm across 15 distinct scenarios, we identify the conditions where the **Opportunity Cost Greedy (OCG)** algorithm provides the most significant strategic advantage. Table 11 highlights the top-performing scenarios based on the improvement over the baseline.

Table 11: Key Strategic Scenarios: Maximum Improvement of OCG over DG.

| Factor                          | Scenario ID | Environment Condition            | Improvement (%) |
|---------------------------------|-------------|----------------------------------|-----------------|
| <b>Points Dist. (Factor A)</b>  | S3          | Star-Heavy Market                | <b>+6.10%</b>   |
| <b>Positional (Factor B)</b>    | S6          | Positional Rigidity              | <b>+4.42%</b>   |
| <b>Market Noise (Factor C)</b>  | S10         | Chaotic Market ( $\delta = +5$ ) | <b>+4.16%</b>   |
| <b>Load/Scarcity (Factor D)</b> | S13         | Extreme Scarcity ( $D = 1$ )     | <b>+6.82%</b>   |

## Analysis of the Competitive Edge:

- **Securing Elite Talent (S3):** In star-heavy drafts, the "cost of waiting" for one more turn is mathematically massive. OCG quantifies this cost, ensuring elite assets are secured before the value "cliff" occurs.
- **Single Position (S6):** When positional flexibility is eliminated, early greedy mistakes are impossible to patch. OCG's foresight prevents these catastrophic positional deficits, maintaining high optimality.
- **Resilience to Market Noise (S10):** Even when opponents' behavior is unpredictable, OCG maintains a robust lead, proving its delay-cost heuristic acts as a natural hedge against market volatility.
- **Navigation of Scarcity (S13):** In environments where almost every player is drafted ( $D = 1$ ), DG often suffers from "positional blindness," leading to a 6.82% loss in roster value. OCG's foresight prevents these late-draft collapses.

## 7 Conclusions and Business Recommendations

For an MLB front office, a successful draft dictates the trajectory of the franchise. While internal scouting evaluations have largely converged across the league, draft execution remains a highly exploitable inefficiency.

This project successfully translated the chaotic, interdependent environment of a fantasy snake draft into a rigorous Operations Research model. We demonstrated that relying on naive "best player available" mentalities (Direct Greedy) bleeds significant roster value, especially when the talent pool is top-heavy or positional flexibility is low.

While the exact ADP-aware ILP provides flawless rosters, the limitless nature of the baseball player pool renders exact optimization operationally dead-on-arrival due to exponential runtime scaling. Our proposed **Opportunity Cost Greedy (OCG)** algorithm brilliantly bridges this gap. By calculating the delay-cost of waiting just one pick into the future, OCG captures the strategic essence of the ILP. It consistently minimizes the optimality gap under severe market stress and processes 50-million variable equivalent draft instances in less than 25 seconds.

**Limitations & Future Extensions:** ADP is only a proxy for opponent behavior; human managers exhibit biases (e.g., hometown favorites) that ADP misses. Furthermore, point projections carry standard deviations (injury risk, slump risk) not currently modeled. Future iterations of this tool could incorporate probabilistic opponent modeling and Monte Carlo simulations on projection variance, upgrading OCG from a deterministic solver to a fully probabilistic draft-day AI.

Ultimately, this report proves to the front office that computational draft optimization is not just a theoretical exercise, but a scalable, real-time weapon capable of maximizing roster value on draft day.